Prediction Intervals for Regression:
Theory, Methods, and Applications

Master Thesis Defense

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Motivation

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- Mortgage lending decisions
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A point estimate is not enough. Decision-making requires uncertainty quantification.

Outline

1. Prediction intervals
2. The fundamental challenge
3. How existing methods address the problem
4. Empirical results
5. Housing price application
6. Main conclusions

What is a prediction interval

Construct an interval $C(X_{n+1})$ such that

$$P(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

Example:

Point estimate

$$\hat{y} = 5.0$$

Prediction interval

$$[4.4,\ 5.6]$$

What do we want?

Validity

The interval should contain the future observation with probability at least $1 - \alpha$.

Efficiency

The interval should be informative and therefore as short as possible.

Adaptivity

The interval width should vary with the level of uncertainty.

Can we achieve all three simultaneously?

What does it mean to be valid?

Marginal Validity

$$P(Y \in C(X)) \geq 1 - \alpha$$

Guarantee on average.

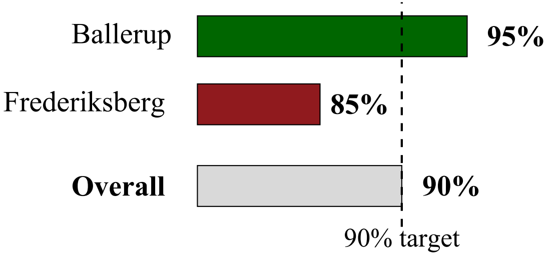
Conditional Validity

$$P(Y \in C(X) \mid X = x) \geq 1 - \alpha$$

Guarantee for every x.

These are fundamentally different.

Why marginal validity may be misleading



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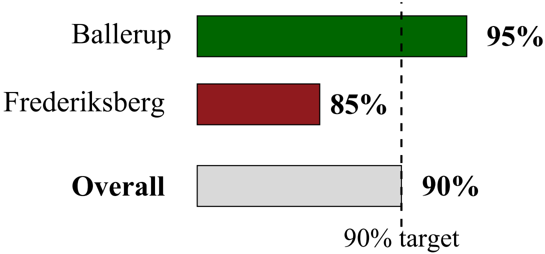
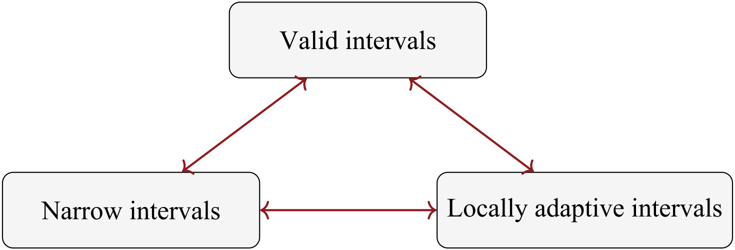


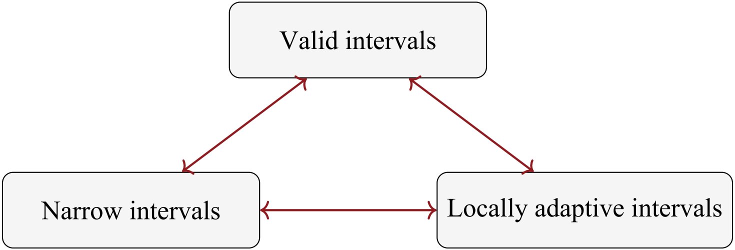
Good average performance

does not imply good local performance.

The fundamental challenge



The fundamental challenge



Cannot generally be achieved simultaneously

Three approaches to the problem

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Gaussian Prediction Intervals

Strong assumptions in exchange for strong theoretical guarantees.

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Quantile Regression

Flexible estimation of conditional quantiles and adaptive interval widths.

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Conformal Prediction

Distribution-free validity under minimal assumptions.

Classical Gaussian prediction intervals

Assume $Y = X^\top \beta + \varepsilon$ with $\varepsilon \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$ and $\varepsilon \perp\!\!\!\perp X$

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Strengths

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- Conditional coverage
- Statistically efficient

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Weaknesses

- Linearity
- Homoskedasticity
- Gaussian errors

Quantile Regression

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- Allows asymmetric intervals
- Flexible extension via QRF

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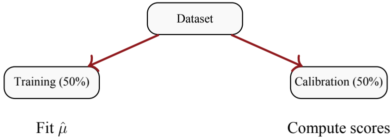
Weaknesses

- Coverage depends on model quality
- No finite-sample guarantee

(Split) Conformal Prediction



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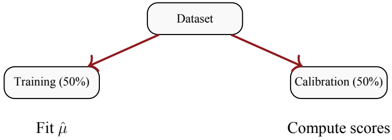

Key guarantee:

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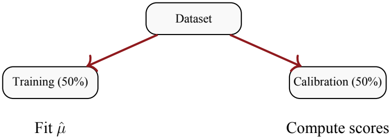

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- Finite-sample validity
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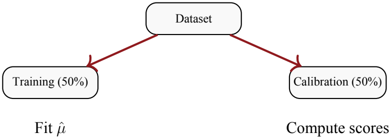

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Strengths

- Finite-sample validity
- Distribution-free
- Model agnostic

Weaknesses

- Marginal validity only
- Limited local adaptivity
- Efficiency depends on model quality

How stable is conformal coverage?

For split conformal prediction, the training-conditional coverage

$$P(Y_{n+1} \in C(X_{n+1}) \mid D_n)$$

is random.

Under suitable assumptions,

$$P(Y_{n+1} \in C(X_{n+1}) \mid D_n) \sim \mathrm{Beta}\big((1-\alpha)(n+1), \alpha(n+1)\big)$$

Consequences:

- Mean $= 1 - \alpha$
- Variance $= \dfrac{\alpha(1-\alpha)}{n+2}$
- Coverage becomes more stable as n increases

The tradeoff

Method	**Marg. Validity**	**Cond. Validity**	**Dist.-free**	**Adaptive Width**	**Finite Sample**
Gaussian PI	(✓)	(✓)	×	×	(✓)
QR	≈	≈	×	✓	×
CP	✓	×	✓	×	✓
Mondrian CP	✓	×	✓	≈	✓
CQR	✓	×	✓	✓	✓

✓ = guaranteed property (✓) = guaranteed property under model assumptions
× = not guaranteed ≈ = holds asymptotically or only approximately

Simulation study

What happens when assumptions fail?

Scenarios:

- Linear vs nonlinear regression
- Symmetric vs asymmetric errors
- Homoskedastic vs heteroskedastic noise

Methods compared:
Gaussian, QR, RQ, CP, and CQR.

Underlying prediction models:
Linear model and random forests

$$n \in \{20, 80, 320, 1280\}, \quad M = 200 \text{ repetitions}, \quad n_{test} = 100,000, \quad 1 - \alpha = 0.9$$

Four experiments, four challenges

Exp. 1: Baseline

$Y = \beta X + a + \varepsilon,\ \varepsilon \sim N(0, \sigma^2)$
All assumptions hold.

Exp. 2: Asymmetric noise

$\varepsilon \sim \mathrm{Exp}(1/\sigma) - \sigma$
Tests symmetry assumptions.

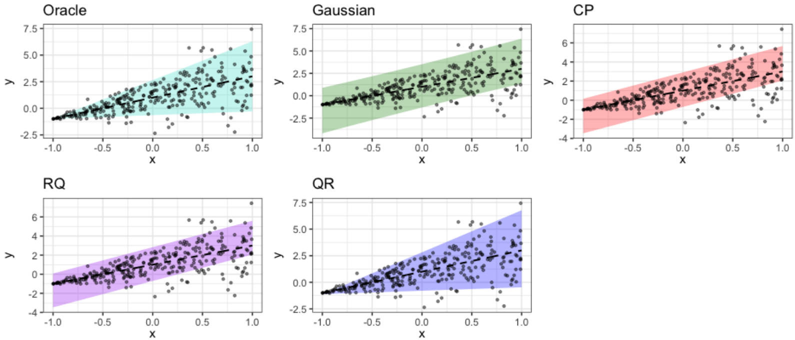
Exp. 3: Nonlinear mean

$\mu(x) = \sin(2\pi x),\ \varepsilon \sim N(0, \sigma^2)$
Tests model flexibility.

Exp. 4: Heteroskedastic

$\sigma(x) = 1 + x$
Tests local adaptivity.

Why global calibration fails locally



- Oracle width grows linearly with x - Gaussian, CP, and RQ stay flat
- Flat intervals overcover for small x, undercover for large x
- QR adapts in width

Main results

Gaussian
performs well when
assumptions hold

Conformal
preserves coverage
across all settings

QR / QRF
improve adaptivity,
less stable coverage

No method dominates across all settings.
Performance depends on the data-generating process

Heatmap over results - with CQR

Greener = closer to nominal coverage/oracle length.
No method is green everywhere.

The limitations

- Gaussian intervals fail under misspecification
- Quantile methods fail when quantiles are poorly estimated
- Conformal intervals may miscover locally

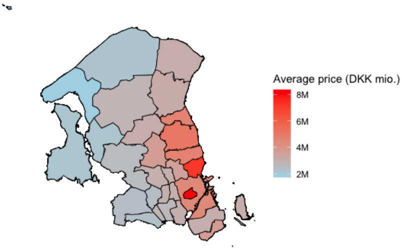
Case study: Housing prices

Data:

- 227,515 housing transactions
- Capital Region of Denmark, 2000-2025
- Highly heterogeneous market

Challenges:

- Nonlinearity
- Heteroskedasticity
- Local variation in uncertainty



Average price (DKK mio.) across municipalities

Case study: summary across methods

Average marginal coverage and interval width, $B = 1000$ repeated splits.

Method	**Coverage**	**Width (mio. DKK)**
Gaussian (LM)	0.911	3.7
Global CP (RF)	0.900	2.6
Mondrian CP (area) (RF)	0.900	2.6
QR (RF)	0.950	3.5
Nominal target	*0.900*	-

- All methods are valid on average
- QR overcovers, paying for its local stability with wider intervals
- CP remains narrowest, despite weaker local adaptivity

Main lessons

1. Validity is not a single concept
2. Conditional validity is fundamentally difficult
3. No method simultaneously optimizes validity, efficiency and adaptivity
4. Method choice depends on the objective

Limitations & future work

Limitations

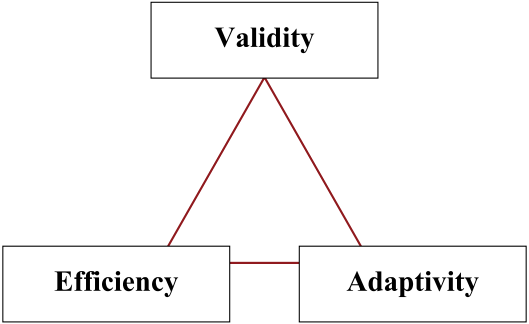
- QRF performance is sensitive to tuning
- Same train/calibration split used across all methods, for comparability

Future work

- Other base learners: boosting, neural nets
- High-dimensional settings
- Cross-validation methods
- Adaptive train/calibration split ratios

Conclusion

The construction of prediction intervals is fundamentally a trade-off problem.



Thank you for listening!

Data splitting

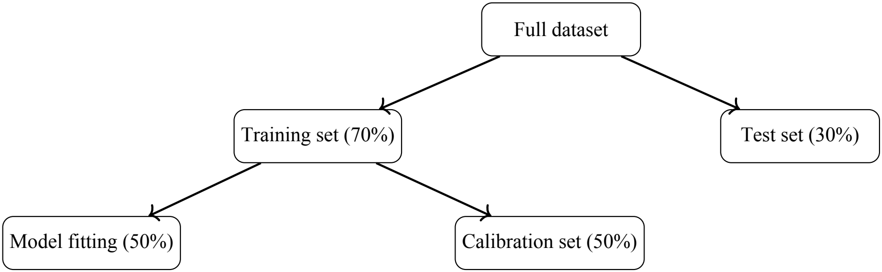
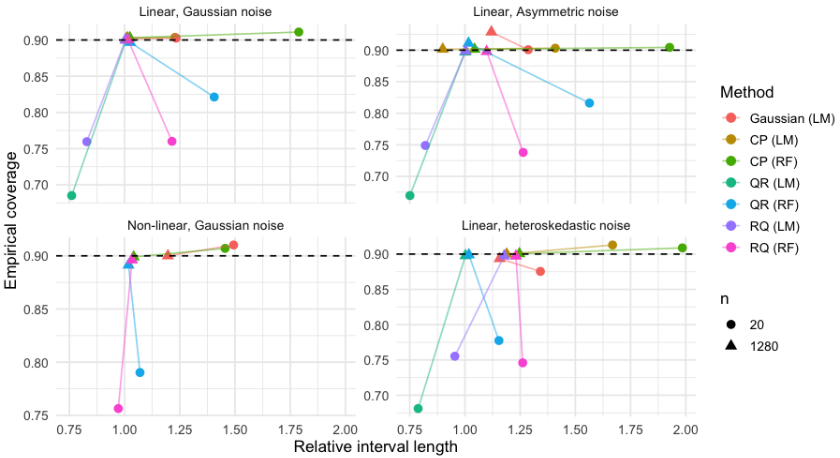


Figure: Data splitting procedure used for split conformal prediction.

Results


Linear, Gaussian noise
Linear, Asymmetric noise
Non-linear, Gaussian noise
Linear, heteroskedastic noise
Empirical coverage
Relative interval length
0.70
0.75
0.80
0.85
0.90
0.75
1.00
1.25
1.50
1.75
2.00
Method
Gaussian (LM)
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n
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